NORP: County-Level Pollution, Health, and Socioeconomic Data Integration for Nonprofit Research

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Introduction

The Nonprofit Organization Research Panel (NORP) is an interdisciplinary, collaborative initiative between the students and faculty of the Computer Science department of Georgia Tech and non-profit research groups from George Mason University, American University, and Urban Institute. The main objective of the initiative is to integrate data that can assist with nonprofit research and cleanly provide a way to analyze and visualize the data. NORP primarily uses a platform called Metabase, which allows researchers to store and consolidate a variety of data relating to nonprofit efforts such as socioeconomic data from the United States Census, IRS990, etc [1]. Metabase allows users to perform SQL and LLM-assisted data querying and create visualizations in order to perform analysis on the data and draw conclusions.

Motivation

Across the United States, there are more than a million nonprofit organizations that operate to support critical needs such as healthcare, environmental protection, education, and community well-being. However, despite their widespread presence, nonprofits face significant limitations in accessing and utilizing data that could guide their strategic decisions, especially in the context of resource allocation. Existing nonprofit research tools are often costly, fragmented, or lack robust infrastructure for integrating, analyzing, and visualizing large-scale datasets. These roadblocks make it difficult for nonprofits to identify where needs are greatest, understand the underlying drivers of community vulnerability, and allocate their efforts effectively. NORP seeks to address these challenges by providing an accessible, centralized platform for nonprofit-relevant data and enabling data-driven research that is both actionable and scalable.

For example, despite the growing urgency surrounding environmental and public health challenges in the United States, nonprofit organizations often lack consolidated and actionable data to guide their decision-making. While NORP includes a diverse range of

datasets that address many issues nonprofits face, there is still a gap in county-level datasets that thoroughly address communities experiencing high pollution levels, including elevated PM2.5 concentrations and poor Air Quality Index (AQI) scores in addition to migration rates in and out of counties and datasets relating to health outcomes and resources. These combined issues can worsen chronic disease prevalence, reduce quality of life, and contribute to population decline as residents migrate away from environmentally disadvantaged areas.

Although agencies such as the EPA, CDC, Census Bureau, and IRS publish environmental, health, and demographic datasets, these resources are typically siloed and difficult to integrate. As a result, organizations seeking to address environmental justice, strengthen community health, or reduce socioeconomic disparities often operate without a clear understanding of where the greatest needs exist or what factors contribute most to risk within a community.

Our work seeks to address that gap by integrating and analyzing multiple complementary datasets, including AQI measures, PM2.5 concentration data, county-level migration flows from IRS records, median household income, poverty rates, and health outcome indicators from the CDC. By examining how environmental conditions relate to economic vulnerability, health burdens, and population movement, the project aims to identify counties that face the highest combined risks. These findings support the broader mission of NORP by offering nonprofits accessible, data-informed insights to guide intervention strategies and resource allocation.

Related Work

Previous research bodies have examined the connections between air pollution, socioeconomic inequality, and population movement across the United States. De Silva et al. (2024) provide one of the most recent analyses by investigating how households relocate in response to pollution exposure [2]. Their study finds that PM2.5 concentrations are correlated with the income levels of out-migrants, indicating that

higher-income households are more likely to move away from areas with elevated pollution levels. This dynamic of selective migration aligns closely with our project's interest in understanding how environmental conditions influence population shifts at the county level.

One study published in the National Library of Medicine composed by Jbaily et al. (2022) analyzes PM2.5 disparities across census tracts and income groups and shows that lower-income communities are consistently exposed to higher pollution levels compared with wealthier populations [3]. Their work emphasizes the importance of integrating pollution and socioeconomic indicators when assessing community vulnerability. These findings support one of the core goals of our project, which is to investigate whether similar disparities are observable at the county-level.

Our project builds on this foundation by integrating multiple datasets across these domains into a unified framework suitable for nonprofit decision-making. Thus, we aim to extend prior research by translating academic findings into a practical data integration resource tailored for use in the Metabase NORP platform.

Datasets

Our project focuses on integrating six different types of data into the NORP database. Each type of data is populated with many datasets that span a varying number of years, making up a total of 36 datasets (counted by the amount of CSV files). Specifically, these datasets all focus on different measures at the county level of the United States.

The main two datasets that drive our project's motivation span over a 10-year period of 2015-2025 for better breadth of data. For air pollution measurements, we selected two datasets from the United States Environmental Protection Agency (EPA) [4]. One dataset category focuses on the Air Quality Index (AQI), giving us measures of Max and Median AQI for counties within the United States and the amount of healthy/unhealthy days experienced annually as determined by the AQI. This data allows us to collect a general measure for the air quality in different counties to allow for analysis of air

pollution. The other dataset category focuses on the concentrations of different monitored substances such as Ozone, Average Ambient Temperature, PM2.5, Solar Radiation, Calcium PM2.5 LC, Iron PM2.5, etc. For our purposes, we extracted Ozone and PM2.5 levels in order to analyze air pollution rates within counties. We chose to focus on these two parameters because they are most relevant to air pollution. Most notably, PM2.5 refers to particulate matter or fine inhalable particles with diameters that are generally 2.5 micrometers and smaller and can cause health issues [5].

In addition, we used migration data from the IRS from 2015-2022 [6], as later years are not available. This data comes in two parts, inflow and outflow, measuring the number of households that leave a county and enter a county. We will use this data to analyze the rates at which households may leave areas due to pollution (using an Inflow - Outflow calculation). In addition, this data also includes AGI (average gross income) which we planned to use for rough income analysis relative to migration and pollution.

In order to better analyze socioeconomic factors in relation to pollution, we also elected to include two datasets that will give us a general measure of the poverty rates and median household incomes. These two datasets come from the U.S Department of Agriculture's Economic Research Service [7] and specifically includes information for poverty estimates of United States' counties in 2023, and median household income by county in 2022. Although our other datasets span a longer range of time, we wanted to use these datasets coupled with AGI from the migration datasets for baseline estimates of socioeconomic status of counties.

We have also included county-level health outcomes data sourced from the CDC PLACES datasets from 2020-2024 [8]. The NORP Metabase currently includes a limited amount of datasets related to health outcome data. Thus, we wanted to integrate health data to analyze trends relating to pollution and enrich the current NORP database as a result to give nonprofits more information. These datasets include information regarding rates of asthma, stroke, diabetes, etc. at the county level.

Approach

Cleaning and Integration

After consolidating our chosen datasets, we needed to carefully inspect the data to determine the most important parameters needed for our planned analysis. Due to the large amount of rows contained in most of the datasets, this step required us to pre-plan the factors we wanted to analyze so that we could write concise code outlining the specific rows to keep and delete. Once we looked through each dataset category and determined the most important information, we created Python scripts using Pandas to clean the data. The scripts involved removing empty/null spaces, special characters, and rows/columns that were irrelevant for our project. One example is shown below:

```python
import pandas as pd
import sys
import os

def clean_aqi_data(year, output_dir):
    """Clean annual AQI by county data for the given year."""
    input_file = os.path.join("aqi_by_county", f"annual_aqi_by_county_{year}.csv")
    output_file = os.path.join(output_dir, f"cleaned_aqi_{year}.csv")

    if not os.path.exists(input_file):
        print(f"[!] File not found for {year}: {input_file}")
        return

    # Load raw dataset
    aqi = pd.read_csv(input_file)

    # Keep only relevant columns
    aqi_clean = aqi[[
        "State",
        "County",
        "Year",
        "Days with AQI",
        "Good Days",
        "Moderate Days",
        "Unhealthy for Sensitive Groups Days",
        "Unhealthy Days",
        "Max AQI",
        "Median AQI"
    ]].copy()

    # Derived metrics
    aqi_clean["Pct_Good"] = aqi_clean["Good Days"] / aqi_clean["Days with AQI"]
    aqi_clean["Pct_Unhealthy"] = (
        (aqi_clean["Unhealthy for Sensitive Groups Days"] + aqi_clean["Unhealthy Days"])
        / aqi_clean["Days with AQI"]
    )

    # Clean up names
    aqi_clean["County"] = aqi_clean["County"].str.replace(" County", "", regex=False).str.strip().str.title()
    aqi_clean["State"] = aqi_clean["State"].str.title()

    # Save cleaned dataset
    aqi_clean.to_csv(output_file, index=False)

    print(f"[✓] Cleaned AQI dataset saved: {output_file}")


if __name__ == "__main__":
    # Create output folder if it doesn't exist
    output_dir = "cleaned_aqi_data"
    os.makedirs(output_dir, exist_ok=True)

    if len(sys.argv) == 2:
        # Run for a specific year
        year = sys.argv[1]
        clean_aqi_data(year, output_dir)
    else:
        # Run for all years 2015-2025
        for year in range(2015, 2026):
            clean_aqi_data(year, output_dir)

        print("\nAll years processed and saved in 'cleaned_aqi_data/'")
```

After creating and running our python scripts for each dataset category, we were left with our finalized cleaned datasets to be integrated into Metabase. The next step was using SQL Create Table statements in order to properly integrate our now cleaned datasets to the existing database. Due to our datasets spanning multiple years for the AQI, migration, and air quality concentration categories, we included a table for each year. One example snippet of the SQL Create Table statements is given below:

```sql
CREATE TABLE AQI_Data_2015 (
    `State` VARCHAR(100),
    `County` VARCHAR(100),
    `Year` INT,
    `Days_with_AQI` INT,
    `Good_Days` INT,
    `Moderate_Days` INT,
    `Unhealthy_Sensitive_Days` INT,
    `Unhealthy_Days` INT,
    `Max_AQI` INT,
    `Median_AQI` INT,
    `Pct_Good` DECIMAL(10,6),
    `Pct_Unhealthy` DECIMAL(10,6)
);

CREATE TABLE AQI_Data_2016 (
    `State` VARCHAR(100),
    `County` VARCHAR(100),
    `Year` INT,
    `Days_with_AQI` INT,
    `Good_Days` INT,
    `Moderate_Days` INT,
    `Unhealthy_Sensitive_Days` INT,
    `Unhealthy_Days` INT,
    `Max_AQI` INT,
    `Median_AQI` INT,
    `Pct_Good` DECIMAL(10,6),
    `Pct_Unhealthy` DECIMAL(10,6)
);

CREATE TABLE AQI_Data_2017 (
    `State` VARCHAR(100),
    `County` VARCHAR(100),
    `Year` INT,
    `Days_with_AQI` INT,
    `Good_Days` INT,
    `Moderate_Days` INT,
    `Unhealthy_Sensitive_Days` INT,
    `Unhealthy_Days` INT,
    `Max_AQI` INT,
    `Median_AQI` INT,
    `Pct_Good` DECIMAL(10,6),
    `Pct_Unhealthy` DECIMAL(10,6)
);
```

After we created our cleaning scripts and SQL create table statements, we adjusted existing scripts create_tables.py and insert_into_tables.py to ensure proper integration of our new datasets and pushed our changes to the NORP GitHub.

Analysis

After integrating AQI, pollutant concentration, socioeconomic, health, and migration datasets into Metabase, we conducted a multi-stage analysis designed to understand where environmental risks overlap with socioeconomic disadvantage and population movement. Each analysis stage began by identifying a key question, then using SQL queries and visualizations to uncover patterns meaningful to nonprofit organizations.

We divided our analysis into four components:

1. **Pollution Burden (Cause)** – Understanding the scale of PM2.5 and AQI risk.
2. **Socioeconomic Vulnerability (Possible Factors)** – Examining whether pollution aligns with income and poverty disparities.
3. **Health Impacts (Effects)** – Linking pollution severity to chronic health outcomes.
4. **Migration Response (Effects)** – Determining whether residents are leaving polluted counties and/or moving into cleaner counties.
5. **Environmental Risk Index (Call to Action)** – Building an index that nonprofits can use to prioritize intervention areas.

These components would make up our Metabase dashboard, separated into 5 different tabs that include the visualizations and analyses corresponding to the specified topic.

1. Establishing Analytical Foundations

Before constructing any queries, we examined the structure and limitations of each dataset. Since we were attempting to analyze at a county level, we wanted to organize counties in our analyses through their unique Federal Information Processing Standards (FIPS) codes. While most of the datasets included FIPS codes, AQI did not include county FIPS. In addition, the AQI and pollutant concentration tables varied by year, and the migration tables required normalization because migration years were presented as ranges (e.g., “2015–2016”).

The first issue we had to reconcile was concerning the ranges of the years. For example, the AQI and air quality concentration data spanned the years of 2015–2025, median household income referred solely to 2022 data, poverty solely to 2023, CDC PLACES health outcome data from 2020–2024, and IRS migration from 2015–2022. We ultimately decided to use the AQI and air quality concentration data as the backbone for breadth of data and utilize the poverty and median household income data as rough measures of the general recent income/socioeconomic level of counties within the United States. Additionally, IRS migration would also be analyzed from 2015–2022, and we would use the health outcome data from 2020–2024 as a baseline measure for recent health outcomes of counties within the United States.

To ensure compatibility across all datasets, we made the following adjustments:

- Standardized county identifiers by creating normalized **state** and **county** columns where FIPS codes were not available in our SQL queries for AQI + Migration
- Verified that all FIPS codes were consistently formatted as 5-digit strings.
- Aligned migration data to the nearest available AQI year to allow multi-year joins.

These preparatory considerations ensured that our SQL queries functioned properly and that our analysis remained organized, consistent, and robust despite the varying data year ranges across datasets.

2. Pollution Burden Queries

To understand baseline environmental exposure, we constructed SQL queries that aggregated PM2.5 concentration and AQI metrics by county and year. During this stage, the focus was on:

- Extracting annual pollutant averages (Arithmetic Mean for PM2.5).
- Computing aggregated AQI metrics such as Median AQI and Percent Unhealthy Days.

Where pollution datasets spanned multiple years, we created year-inclusive queries to support long-horizon comparisons (e.g., 2015–2025 PM2.5 averages). These queries formed the foundation for identifying high-burden counties and linking pollution severity to other factors in subsequent steps

For example, some of the queries for creating our visualizations aggregating the AQI and Air Quality Concentration Pollutant data is shown below:

```sql
WITH county_pollution AS (
    SELECT
        c.county_fips,
        c.`County Name`,
        c.`State Name`,
        c.Year,
        ROUND(AVG(c.`Arithmetic Mean`), 2) AS avg_pm25_concentration,
        ROUND(AVG(a.Median_AQI), 2)        AS avg_median_aqi,
        ROUND(AVG(a.Unhealthy_Days), 2)    AS avg_unhealthy_days,
        ROUND(AVG(a.Pct_Unhealthy), 2)     AS avg_pct_unhealthy
    FROM (
        -- PM2.5 tables (add Year manually)
        SELECT *, 2015 AS Year FROM Air_Quality_Concentration_2015
        UNION ALL SELECT *, 2016 AS Year FROM Air_Quality_Concentration_2016
        UNION ALL SELECT *, 2017 AS Year FROM Air_Quality_Concentration_2017
        UNION ALL SELECT *, 2018 AS Year FROM Air_Quality_Concentration_2018
        UNION ALL SELECT *, 2019 AS Year FROM Air_Quality_Concentration_2019
        UNION ALL SELECT *, 2020 AS Year FROM Air_Quality_Concentration_2020
        UNION ALL SELECT *, 2021 AS Year FROM Air_Quality_Concentration_2021
        UNION ALL SELECT *, 2022 AS Year FROM Air_Quality_Concentration_2022
        UNION ALL SELECT *, 2023 AS Year FROM Air_Quality_Concentration_2023
        UNION ALL SELECT *, 2024 AS Year FROM Air_Quality_Concentration_2024
        UNION ALL SELECT *, 2025 AS Year FROM Air_Quality_Concentration_2025
    ) c
    LEFT JOIN (
        SELECT * FROM AQI_Data_2015
        UNION ALL SELECT * FROM AQI_Data_2016
        UNION ALL SELECT * FROM AQI_Data_2017
        UNION ALL SELECT * FROM AQI_Data_2018
        UNION ALL SELECT * FROM AQI_Data_2019
        UNION ALL SELECT * FROM AQI_Data_2020
        UNION ALL SELECT * FROM AQI_Data_2021
        UNION ALL SELECT * FROM AQI_Data_2022
        UNION ALL SELECT * FROM AQI_Data_2023
        UNION ALL SELECT * FROM AQI_Data_2024
        UNION ALL SELECT * FROM AQI_Data_2025
    ) a
      ON c.`County Name`  = a.County
     AND c.`State Name`   = a.State
     AND c.Year = a.Year
    WHERE c.`Parameter Name` = 'PM2.5 - Local Conditions'
    GROUP BY c.county_fips, c.`County Name`, c.`State Name`, c.Year
)

SELECT
    county_fips,
    `County Name`,
    `State Name`,
    Year,
    avg_pm25_concentration,
    avg_median_aqi,
    avg_unhealthy_days,
    avg_pct_unhealthy
FROM county_pollution
WHERE
    avg_pm25_concentration IS NOT NULL
    AND avg_median_aqi      IS NOT NULL
    AND avg_unhealthy_days  IS NOT NULL
    AND avg_pct_unhealthy   IS NOT NULL
ORDER BY avg_pm25_concentration DESC
LIMIT 100;
```

```sql
SELECT
    a.Year,
    ROUND(AVG(c.`Arithmetic Mean`), 2) AS avg_pm25_concentration,
    ROUND(AVG(a.Median_AQI), 2) AS avg_median_aqi
FROM (
    SELECT * FROM Air_Quality_Concentration_2015
    UNION ALL SELECT * FROM Air_Quality_Concentration_2016
    UNION ALL SELECT * FROM Air_Quality_Concentration_2017
    UNION ALL SELECT * FROM Air_Quality_Concentration_2018
    UNION ALL SELECT * FROM Air_Quality_Concentration_2019
    UNION ALL SELECT * FROM Air_Quality_Concentration_2020
    UNION ALL SELECT * FROM Air_Quality_Concentration_2021
    UNION ALL SELECT * FROM Air_Quality_Concentration_2022
    UNION ALL SELECT * FROM Air_Quality_Concentration_2023
    UNION ALL SELECT * FROM Air_Quality_Concentration_2024
    UNION ALL SELECT * FROM Air_Quality_Concentration_2025
) c
JOIN (
    SELECT * FROM AQI_Data_2015
    UNION ALL SELECT * FROM AQI_Data_2016
    UNION ALL SELECT * FROM AQI_Data_2017
    UNION ALL SELECT * FROM AQI_Data_2018
    UNION ALL SELECT * FROM AQI_Data_2019
    UNION ALL SELECT * FROM AQI_Data_2020
    UNION ALL SELECT * FROM AQI_Data_2021
    UNION ALL SELECT * FROM AQI_Data_2022
    UNION ALL SELECT * FROM AQI_Data_2023
    UNION ALL SELECT * FROM AQI_Data_2024
    UNION ALL SELECT * FROM AQI_Data_2025
) a
    ON c.`County Name` = a.County AND c.`State Name` = a.State
GROUP BY a.Year
ORDER BY a.Year ASC;
```

```sql
WITH pollution_data AS (
    SELECT
        c.`State Name`,
        c.Year,
        AVG(c.`Arithmetic Mean`) AS avg_pm25_concentration,
        AVG(a.Median_AQI) AS avg_median_aqi
    FROM (
        SELECT *, 2015 AS Year FROM Air_Quality_Concentration_2015
        UNION ALL SELECT *, 2016 AS Year FROM Air_Quality_Concentration_2016
        UNION ALL SELECT *, 2017 AS Year FROM Air_Quality_Concentration_2017
        UNION ALL SELECT *, 2018 AS Year FROM Air_Quality_Concentration_2018
        UNION ALL SELECT *, 2019 AS Year FROM Air_Quality_Concentration_2019
        UNION ALL SELECT *, 2020 AS Year FROM Air_Quality_Concentration_2020
        UNION ALL SELECT *, 2021 AS Year FROM Air_Quality_Concentration_2021
        UNION ALL SELECT *, 2022 AS Year FROM Air_Quality_Concentration_2022
        UNION ALL SELECT *, 2023 AS Year FROM Air_Quality_Concentration_2023
        UNION ALL SELECT *, 2024 AS Year FROM Air_Quality_Concentration_2024
        UNION ALL SELECT *, 2025 AS Year FROM Air_Quality_Concentration_2025
    ) c
    LEFT JOIN (
        SELECT * FROM AQI_Data_2015
        UNION ALL SELECT * FROM AQI_Data_2016
        UNION ALL SELECT * FROM AQI_Data_2017
        UNION ALL SELECT * FROM AQI_Data_2018
        UNION ALL SELECT * FROM AQI_Data_2019
        UNION ALL SELECT * FROM AQI_Data_2020
        UNION ALL SELECT * FROM AQI_Data_2021
        UNION ALL SELECT * FROM AQI_Data_2022
        UNION ALL SELECT * FROM AQI_Data_2023
        UNION ALL SELECT * FROM AQI_Data_2024
        UNION ALL SELECT * FROM AQI_Data_2025
    ) a
     ON c.`County Name` = a.County
     AND c.`State Name` = a.State
     AND c.Year = a.Year
    WHERE c.`Parameter Name` = 'PM2.5 - Local Conditions'
    GROUP BY c.`State Name`, c.Year
)

SELECT
    ANY_VALUE(`State Name`) AS `State Name`,
    ROUND(AVG(avg_pm25_concentration), 2) AS avg_pm25_concentration,
    ROUND(AVG(avg_median_aqi), 2) AS avg_median_aqi
FROM pollution_data
GROUP BY `State Name`
ORDER BY avg_pm25_concentration DESC;
```

3. Socioeconomic Queries

The next component included mapping socioeconomic data to pollution in the forms of the established PM2.5 and Median AQI measures. This included combining tables of Median Income from 2022 with average PM2.5 and Median AQI from 2015-2025 and Poverty Rate from 2023 with average PM2.5 from 2015-2025.

Examples of the queries used are shown below:

```sql
SELECT
    mi.FIPS_Code AS county_fips,
    mi.Area_Name AS county_name,
    mi.Median_Income,
    AVG(pm.`Arithmetic Mean`) AS avg_pm25_concentration,
    AVG(aqi.Median_AQI) AS avg_median_aqi
FROM Median_Income_2022 mi
JOIN Air_Quality_Concentration_2023 pm
    ON mi.FIPS_Code = pm.county_fips
JOIN AQI_Data_2023 aqi
    ON mi.Area_Name LIKE CONCAT('%', aqi.County, '%')
GROUP BY mi.FIPS_Code, mi.Area_Name, mi.Median_Income
ORDER BY avg_pm25_concentration DESC;
```

```sql
SELECT
    mi.FIPS_Code AS county_fips,
    mi.Area_Name AS county_name,
    mi.Median_Income,
    AVG(pm.`Arithmetic Mean`) AS avg_pm25_concentration,
    AVG(aqi.Median_AQI) AS avg_median_aqi
FROM Median_Income_2022 mi
JOIN Air_Quality_Concentration_2023 pm
    ON mi.FIPS_Code = pm.county_fips
JOIN AQI_Data_2023 aqi
    ON mi.Area_Name LIKE CONCAT('%', aqi.County, '%')
GROUP BY mi.FIPS_Code, mi.Area_Name, mi.Median_Income
ORDER BY avg_pm25_concentration DESC;
```

```sql
WITH pm25_all_years AS (
    SELECT *, 2015 AS Year FROM Air_Quality_Concentration_2015
    UNION ALL SELECT *, 2016 AS Year FROM Air_Quality_Concentration_2016
    UNION ALL SELECT *, 2017 AS Year FROM Air_Quality_Concentration_2017
    UNION ALL SELECT *, 2018 AS Year FROM Air_Quality_Concentration_2018
    UNION ALL SELECT *, 2019 AS Year FROM Air_Quality_Concentration_2019
    UNION ALL SELECT *, 2020 AS Year FROM Air_Quality_Concentration_2020
    UNION ALL SELECT *, 2021 AS Year FROM Air_Quality_Concentration_2021
    UNION ALL SELECT *, 2022 AS Year FROM Air_Quality_Concentration_2022
    UNION ALL SELECT *, 2023 AS Year FROM Air_Quality_Concentration_2023
    UNION ALL SELECT *, 2024 AS Year FROM Air_Quality_Concentration_2024
    UNION ALL SELECT *, 2025 AS Year FROM Air_Quality_Concentration_2025
)

SELECT
    p.FIPS_Code AS county_fips,
    p.Area_Name,
    pm.Year,
    p.Poverty_Percentage,
    ROUND(AVG(pm.`Arithmetic Mean`), 2) AS avg_pm25
FROM Poverty_Rates_2023 p
JOIN pm25_all_years pm
    ON pm.county_fips = p.FIPS_Code
WHERE pm.`Parameter Name` LIKE '%PM2.5%'
GROUP BY
    p.FIPS_Code,
    p.Area_Name,
    pm.Year,
    p.Poverty_Percentage
ORDER BY avg_pm25 DESC;
```

4. Health Outcome Queries

To explore how environmental pollution aligns with health concerns afflicting communities, we linked chronic disease prevalence data from CDC PLACES (2020–2024) with pollution metrics (PM2.5, Median AQI). These queries focused on answering whether counties with higher pollution levels experienced a higher rate of respiratory disease or health outcomes.

We highlighted four chronic disease indicators: COPD, Asthma, Heart Disease, and Stroke to see their relevance to long-term exposure to fine particulate matter (PM2.5). By combining our CDC Places Health Datasets with our pollution table, we examined counties with a high prevalence of pollution sensitive conditions.

The sample queries consisted of comparing average PM2.5 and AQI Data per county and combining them with the corresponding health outcomes. This allowed for us to showcase scatter plots highlighting trends between each health condition, and create a table that showcases specific counties with both pollution and health risk.

```sql
WITH combined AS (
    SELECT
        pm.county_fips,
        pm.avg_pm25,
        h.copd_prev
    FROM (
        SELECT county_fips, AVG(`Arithmetic Mean`) AS avg_pm25
        FROM (
            SELECT * FROM Air_Quality_Concentration_2020
            UNION ALL SELECT * FROM Air_Quality_Concentration_2021
            UNION ALL SELECT * FROM Air_Quality_Concentration_2022
            UNION ALL SELECT * FROM Air_Quality_Concentration_2023
        ) pm
        GROUP BY county_fips
    ) pm
    JOIN Places_County_Health_2023 h USING (county_fips)
),
averages AS (
    SELECT
        AVG(avg_pm25) AS avg_pm,
        AVG(copd_prev) AS avg_copd
    FROM combined
)
SELECT
    CASE
        WHEN avg_pm25 <= (SELECT avg_pm FROM averages)
             AND copd_prev <= (SELECT avg_copd FROM averages)
            THEN 'Low Pollution / Low Health Risk'
        WHEN avg_pm25 > (SELECT avg_pm FROM averages)
             AND copd_prev <= (SELECT avg_copd FROM averages)
            THEN 'High Pollution / Low Health Risk'
        WHEN avg_pm25 <= (SELECT avg_pm FROM averages)
             AND copd_prev > (SELECT avg_copd FROM averages)
            THEN 'Low Pollution / High Health Risk'
        ELSE 'High Pollution / High Health Risk'
    END AS category,
    COUNT(*) AS county_count
FROM combined
GROUP BY category;
```

```sql
WITH pm25_multi_year AS (
    SELECT
        county_fips,
        AVG(`Arithmetic Mean`) AS avg_pm25
    FROM (
        SELECT * FROM Air_Quality_Concentration_2019
        UNION ALL SELECT * FROM Air_Quality_Concentration_2020
        UNION ALL SELECT * FROM Air_Quality_Concentration_2021
        UNION ALL SELECT * FROM Air_Quality_Concentration_2022
        UNION ALL SELECT * FROM Air_Quality_Concentration_2023
    ) pm
    GROUP BY county_fips
)
SELECT
    pm.avg_pm25,
    h.asthma_prev,
    h.county_name,
    h.state,
    h.pop_total AS population
FROM pm25_multi_year pm
JOIN Places_County_Health_2023 h ON pm.county_fips = h.county_fips
WHERE pm.avg_pm25 BETWEEN 5 AND 15
ORDER BY pm.avg_pm25;
```

```sql
WITH pm25_multi_year AS (
    SELECT
        county_fips,
        `State Name` AS state,
        `County Name` AS county_name,
        AVG(`Arithmetic Mean`) AS avg_pm25
    FROM (
        SELECT * FROM Air_Quality_Concentration_2019
        UNION ALL SELECT * FROM Air_Quality_Concentration_2020
        UNION ALL SELECT * FROM Air_Quality_Concentration_2021
        UNION ALL SELECT * FROM Air_Quality_Concentration_2022
        UNION ALL SELECT * FROM Air_Quality_Concentration_2023
    ) AS pm_all
    GROUP BY county_fips, state, county_name
),
health AS (
    SELECT county_fips, county_name, state, year, copd_prev, pop_total
    FROM Places_County_Health_2023
),
stats AS (
    SELECT
        (SELECT AVG(avg_pm25) FROM pm25_multi_year) AS avg_pm25_mean,
        (SELECT STDDEV(avg_pm25) FROM pm25_multi_year) AS avg_pm25_std,
        (SELECT AVG(copd_prev) FROM health) AS copd_mean,
        (SELECT STDDEV(copd_prev) FROM health) AS copd_std
)
SELECT
    h.county_name,
    h.state,
    h.year,
    ROUND(pm.avg_pm25, 2) AS avg_pm25,
    ROUND(h.copd_prev, 2) AS copd_prev,
    h.pop_total AS population
FROM pm25_multi_year pm
JOIN health h USING (county_fips)
JOIN stats s
ORDER BY ((pm.avg_pm25 - s.avg_pm25_mean) / s.avg_pm25_std) + ((h.copd_prev - s.copd_mean) / s.copd_std) DESC;
```

```sql
WITH pm25_multi_year AS (
    SELECT
        pm.county_fips,
        AVG(pm.`Arithmetic Mean`) AS avg_pm25
    FROM (
        SELECT * FROM Air_Quality_Concentration_2019
        UNION ALL
        SELECT * FROM Air_Quality_Concentration_2020
        UNION ALL
        SELECT * FROM Air_Quality_Concentration_2021
        UNION ALL
        SELECT * FROM Air_Quality_Concentration_2022
        UNION ALL
        SELECT * FROM Air_Quality_Concentration_2023
    ) pm
    GROUP BY pm.county_fips
),
categorized AS (
    SELECT
        m.county_fips,
        m.avg_pm25,
        CASE
            WHEN m.avg_pm25 < 5 THEN '<5 µg/m³'
            WHEN m.avg_pm25 < 10 THEN '5–10 µg/m³'
            ELSE '10–15 µg/m³'
        END AS pollution_risk_level,
        CASE
            WHEN m.avg_pm25 < 5 THEN 1
            WHEN m.avg_pm25 < 10 THEN 2
            ELSE 3
        END AS pollution_order
    FROM pm25_multi_year m
),
joined AS (
    SELECT
        pollution_risk_level,
        AVG(h.asthma_prev) AS avg_asthma,
        AVG(h.copd_prev) AS avg_copd,
        AVG(h.chd_prev) AS avg_chd,
        AVG(h.stroke_prev) AS avg_stroke,
        AVG(h.smoking_prev) AS avg_smoking,
        AVG(h.phlth_prev) AS avg_poor_physical,
        AVG(h.mhlth_prev) AS avg_poor_mental,
        COUNT(*) AS num_counties,
        MIN(pollution_order) AS pollution_order
    FROM categorized c
    JOIN Places_County_Health_2023 h
        ON c.county_fips = h.county_fips
    GROUP BY pollution_risk_level
)
SELECT *
FROM joined
ORDER BY pollution_order ASC;
```

These queries helped us answer whether long term pollution (pm 2.5) exposure is associated with Respiratory and Cardiovascular health outcomes.

5. Migration Response Queries

Migration data analysis required additional preprocessing because although the IRS datasets included county FIPS codes, the naming conventions for county names did not perfectly align with AQI datasets (which did not have county FIPS), and the datasets were reported in multi-year ranges (e.g., “2015–2016”). To support reliable joins with the AQI datasets, we began by creating normalized state and county columns for every migration entry and aligning each migration period with the corresponding AQI years (e.g., 2015-2016 in migration would correspond to 2015 in AQI data). Once the migration data was join-ready, we constructed a scoring system to standardize migration alongside AQI-related metrics. Because migration operates at a dramatically larger numeric scale than daily AQI or pollutant percentages, we applied transformations to ensure all variables contributed proportionally to a unified composite score. Specifically, the migration-AQI scoring queries combined Median AQI, Percent Unhealthy Days, and Net Migration using:

$$(2 \times \text{Median_AQI}) + (100 \times \text{Pct_Unhealthy}) - (\text{net_migration} \div 1000)$$

Median AQI is multiplied by 2 to increase its influence, Percent Unhealthy Days is multiplied by 100 to convert small decimal values into a comparable range, and Net Migration is divided by 1000 so that very large population flows do not dominate the score. These adjustments allow all three metrics to be integrated into one composite ranking that reflects both pollution severity and the magnitude of population outflow. These adjustments ensured that the scoring query produced a balanced composite value that could then be used in further analysis and visualization.

The figures on the following pages are snippets of the SQL queries demonstrating the creation of the normalized state and county columns, as well as the execution of the weighted formula for ordering analysis.

```sql
WITH migration_all_years AS (
    SELECT *, 2015 AS Migration_Year FROM irs_migration_2015_2016
    UNION ALL SELECT *, 2016 AS Migration_Year FROM irs_migration_2016_2017
    UNION ALL SELECT *, 2017 AS Migration_Year FROM irs_migration_2017_2018
    UNION ALL SELECT *, 2018 AS Migration_Year FROM irs_migration_2018_2019
    UNION ALL SELECT *, 2019 AS Migration_Year FROM irs_migration_2019_2020
    UNION ALL SELECT *, 2020 AS Migration_Year FROM irs_migration_2020_2021
    UNION ALL SELECT *, 2021 AS Migration_Year FROM irs_migration_2021_2022
),

/* ------------ NORMALIZE MIGRATION ------------ */
migration_norm AS (
    SELECT
        county_fips,

        /* convert state abbreviation → full lowercase state name */
        CASE
            WHEN state = 'AL' THEN 'alabama'
            WHEN state = 'AK' THEN 'alaska'
            WHEN state = 'AZ' THEN 'arizona'
            WHEN state = 'AR' THEN 'arkansas'
            WHEN state = 'CA' THEN 'california'
            WHEN state = 'CO' THEN 'colorado'
            WHEN state = 'CT' THEN 'connecticut'
            WHEN state = 'DE' THEN 'delaware'
            WHEN state = 'FL' THEN 'florida'
            WHEN state = 'GA' THEN 'georgia'
            WHEN state = 'HI' THEN 'hawaii'
            WHEN state = 'ID' THEN 'idaho'
            WHEN state = 'IL' THEN 'illinois'
            WHEN state = 'IN' THEN 'indiana'
            WHEN state = 'IA' THEN 'iowa'
            WHEN state = 'KS' THEN 'kansas'
            WHEN state = 'KY' THEN 'kentucky'
            WHEN state = 'LA' THEN 'louisiana'
            WHEN state = 'ME' THEN 'maine'
            WHEN state = 'MD' THEN 'maryland'
            WHEN state = 'MA' THEN 'massachusetts'
            WHEN state = 'MI' THEN 'michigan'
            WHEN state = 'MN' THEN 'minnesota'
            WHEN state = 'MS' THEN 'mississippi'
            WHEN state = 'MO' THEN 'missouri'
            WHEN state = 'MT' THEN 'montana'
            WHEN state = 'NE' THEN 'nebraska'
            WHEN state = 'NV' THEN 'nevada'
            WHEN state = 'NH' THEN 'new hampshire'
            WHEN state = 'NJ' THEN 'new jersey'
            WHEN state = 'NM' THEN 'new mexico'
            WHEN state = 'NY' THEN 'new york'
            WHEN state = 'NC' THEN 'north carolina'
            WHEN state = 'ND' THEN 'north dakota'
```

```sql
        END AS state_norm,

        countyname,
        net_migration,
        Migration_Year,

        /* Normalize county names */
        LOWER(
            REGEXP_REPLACE(countyname,
                '( county.*$| parish.*$| total migration.*$)',
            '')
        ) AS county_norm

    FROM migration_all_years
),

/* ------------ NORMALIZE AQI ------------ */
aqi_all_years AS (
    SELECT * FROM AQI_Data_2015
    UNION ALL SELECT * FROM AQI_Data_2016
    UNION ALL SELECT * FROM AQI_Data_2017
    UNION ALL SELECT * FROM AQI_Data_2018
    UNION ALL SELECT * FROM AQI_Data_2019
    UNION ALL SELECT * FROM AQI_Data_2020
    UNION ALL SELECT * FROM AQI_Data_2021
),

aqi_norm AS (
    SELECT
        LOWER(State) AS state_norm,
        LOWER(County) AS county_norm,
        Year,
        Median_AQI,
        Pct_Unhealthy
    FROM aqi_all_years
)

SELECT
    m.county_fips,
    m.state_norm AS state,
    m.countyname,
    a.Year AS pollution_year,
    a.Median_AQI,
    m.net_migration,
    a.Pct_Unhealthy
FROM migration_norm m
JOIN aqi_norm a
      ON m.state_norm = a.state_norm
     AND m.county_norm = a.county_norm
     AND m.Migration_Year = a.Year
ORDER BY
    (a.Median_AQI * 2) +
    (a.Pct_Unhealthy * 100) -
    (m.net_migration / 1000) DESC;
```

6. Composite Environmental Risk Index (ERI) Queries

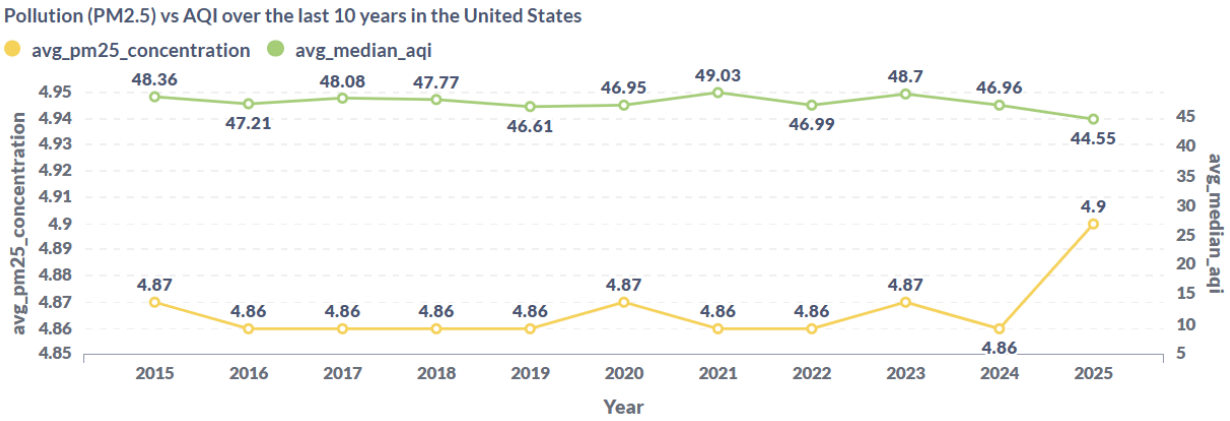
The final stage of the analysis involved constructing a composite Environmental Risk Index (ERI) that integrated pollution, socioeconomic, and health variables into a single weighted metric. The goal of this index was to create a standardized measure that nonprofits could use to identify counties where environmental burden and vulnerability are jointly elevated. To build the ERI, we developed SQL queries that joined PM2.5 averages, median AQI, poverty rates, and COPD (Chronic Obstructive Pulmonary Disease) prevalence at the county level, ensuring all fields were aligned through consistent FIPS codes and matched year ranges. Because each variable operates on a different numerical scale, we incorporated weighting to balance their contributions. The ERI formula assigns 45% weight to PM2.5, 30% to median AQI, 15% to poverty percentage, and 10% to COPD prevalence. This weighting structure emphasizes pollutant exposure while still accounting for socioeconomic and health-related susceptibility. Implementing the ERI required careful handling of missing values, validation of join logic across multiple datasets, and normalization of the final output to ensure comparability across counties.

This index was created to give nonprofit organizations a clear, data-driven tool for pinpointing counties most in need of support, allowing them to prioritize interventions and allocate resources more effectively in areas facing the greatest environmental strain. Using this index, we created a SQL query to display the top 100 counties with the highest ERI scores:

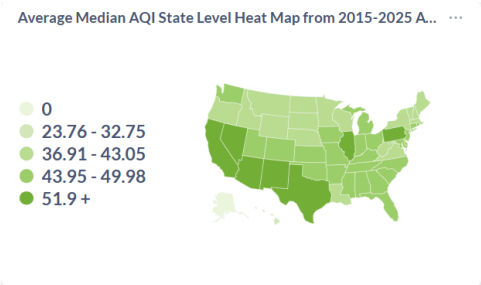
```sql
SELECT
    pm.county_fips,
    pm.`County Name`,
    AVG(pm.`Arithmetic Mean`) AS avg_pm25,
    aq.Median_AQI,
    p.Poverty_Percentage,
    i.Median_Income,
    h.copd_prev,

    -- ERI Calculation
    (
        0.45 * AVG(pm.`Arithmetic Mean`) +
        0.30 * aq.Median_AQI +
        0.15 * p.Poverty_Percentage +
        0.10 * h.copd_prev
    ) AS ERI

FROM Air_Quality_Concentration_2020 pm

JOIN AQI_Data_2020 aq
    ON aq.County = pm.`County Name`
    AND aq.State = pm.`State Name`

JOIN Poverty_Rates_2023 p
    ON p.FIPS_Code = pm.county_fips

JOIN Median_Income_2022 i
    ON i.FIPS_Code = pm.county_fips

JOIN Places_County_Health_2020 h
    ON h.county_fips = pm.county_fips

WHERE pm.`Parameter Name` LIKE '%PM2.5%'

GROUP BY
    pm.county_fips,
    pm.`County Name`,
    aq.Median_AQI,
    p.Poverty_Percentage,
    i.Median_Income,
    h.copd_prev

ORDER BY ERI DESC
LIMIT 100;
```

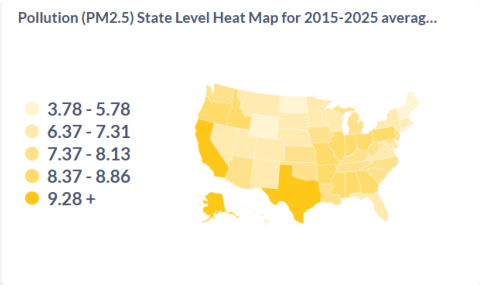
Results

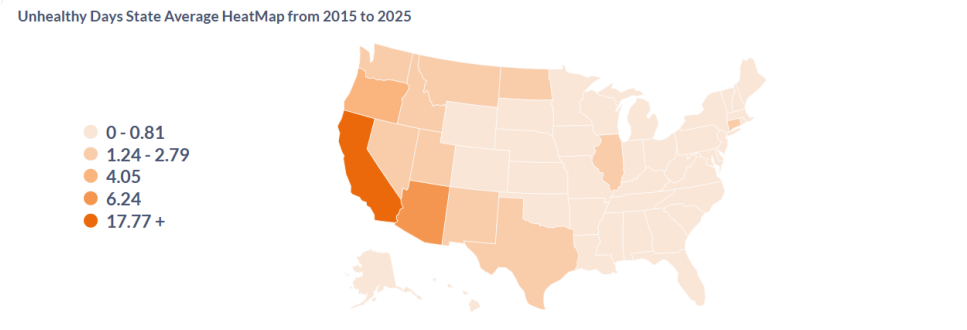
Pollution Severity & Exposure Trends (2015 to 2025)



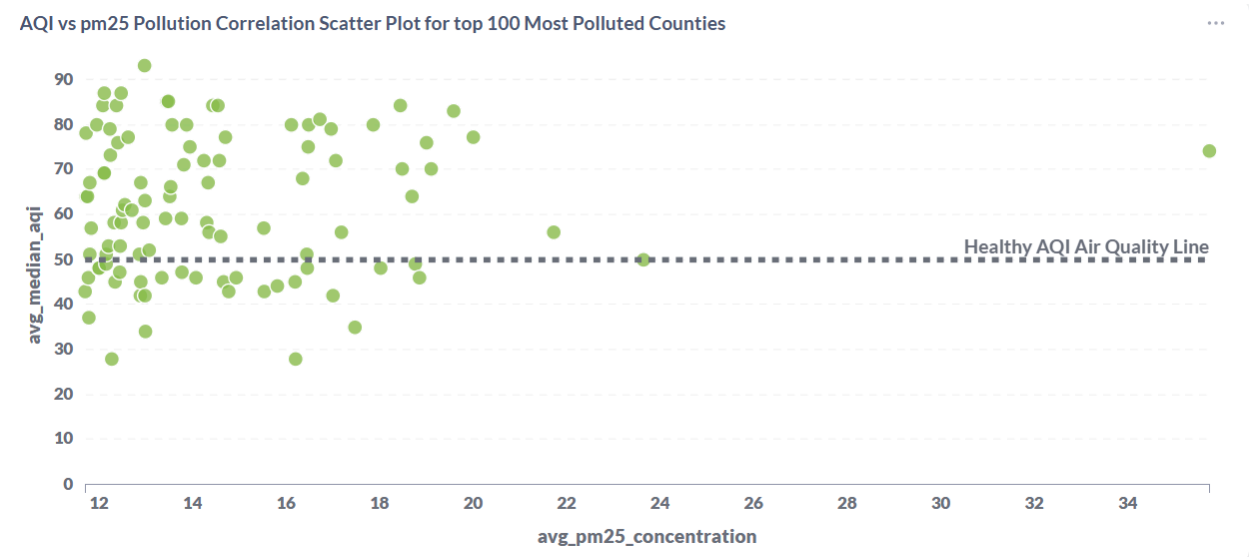
The above line graph shows the average PM 2.5 Concentration and AQI Median Values averaged across every county and from 2015 to 2025 to highlight trends. The PM2.5 levels remain steady throughout the decade but showcase a slight increase in 2025. This can imply a constant exposure to pollution during the last decade which could have adverse effects on health. The AQI shows some fluctuation, with visible peaks in 2021 and 2023 with a recent shift downwards in 2025. This indicated that despite their being more particulate matter in the air the AQI as an average conglomerated over the United States remained stable.





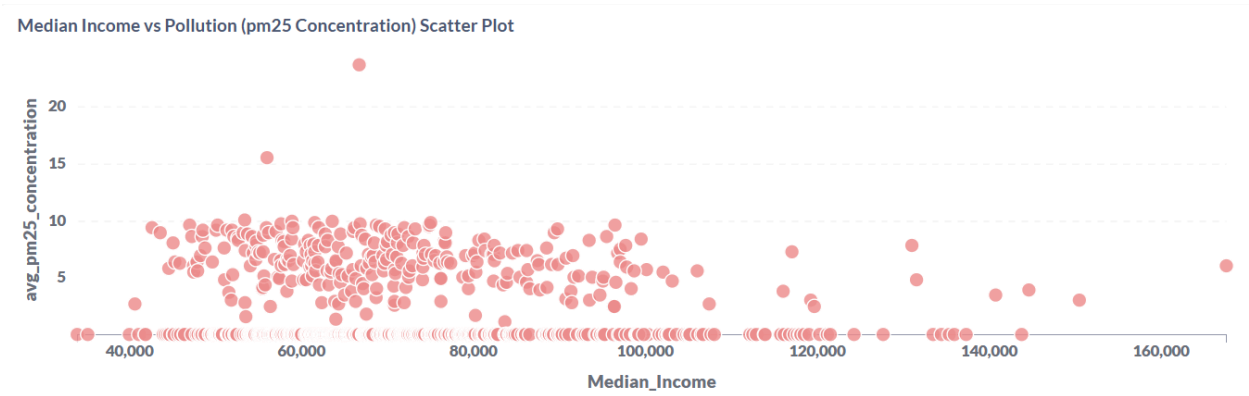


The maps above showcase pollution levels and statistics on a State Level. From these maps, California, New Mexico, Nevada, Texas, and some other states showcase a higher range for PM2.5, AQI, and Unhealthy Air Days. The South Eastern States showcase a higher AQI and PM2.5 Average but not a high range for Unhealthy Days. These patterns aim to highlight regions of the United States that are affected by consistent environmental exposure in the air.



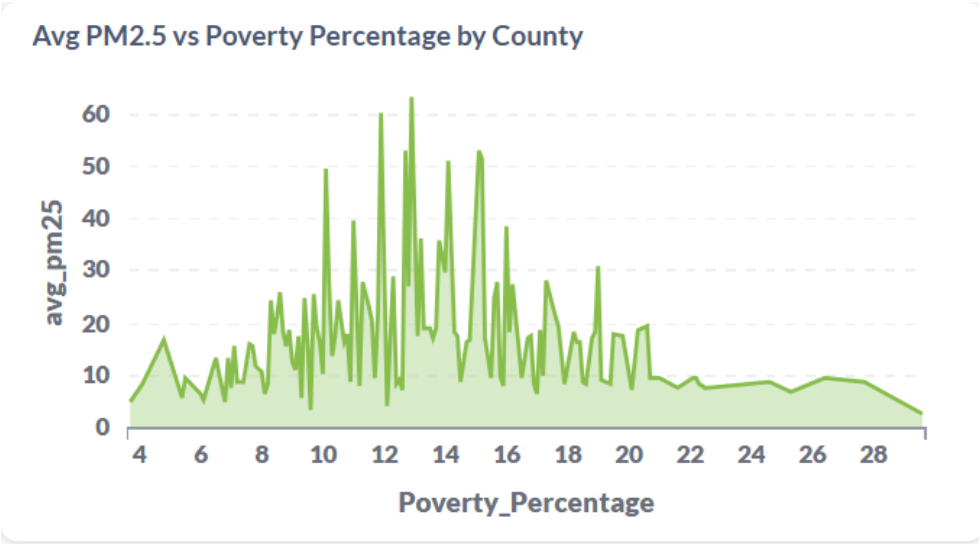
The scatter plot above showcases PM2.5 vs AQI Levels for the 100 counties listed above in the table with the highest PM2.5. While some countries showcased a high AQI despite having a low PM2.5, as PM2.5 increased more counties were found higher than the AQI 50 point threshold, which is an indicator for when Air Quality can be flagged as being unsafe for Sensitive groups. This showcases that AQI and Pm2.5 are not directly correlated, but a higher PM2.5 concentration can more consistently indicate an elevated Air Quality Risk.

Income and Poverty Trends compared with Pollution



The scatter plot above showcases the average PM2.5 levels compared to the median income across counties throughout the United States. The higher pollution level is apparent in both lower and higher income counties showing that there is no consistent trend towards income and pollution. However, this scatterplot also highlights the frequency of counties with both a high PM2.5 level and low Median Income showing that there are communities across the United States that may need some more assistance to help deal with this pollution.

This scatter plot compares AQI (Air Quality Index) vs Median Income across counties in the United States. While AQI values fluctuate above and below the AQI Health Line (50) at the lower income levels, most higher Median Income counties fall below the AQI 50 Threshold. This pattern indicates that counties with a higher median income are more likely to experience cleaner air conditions.

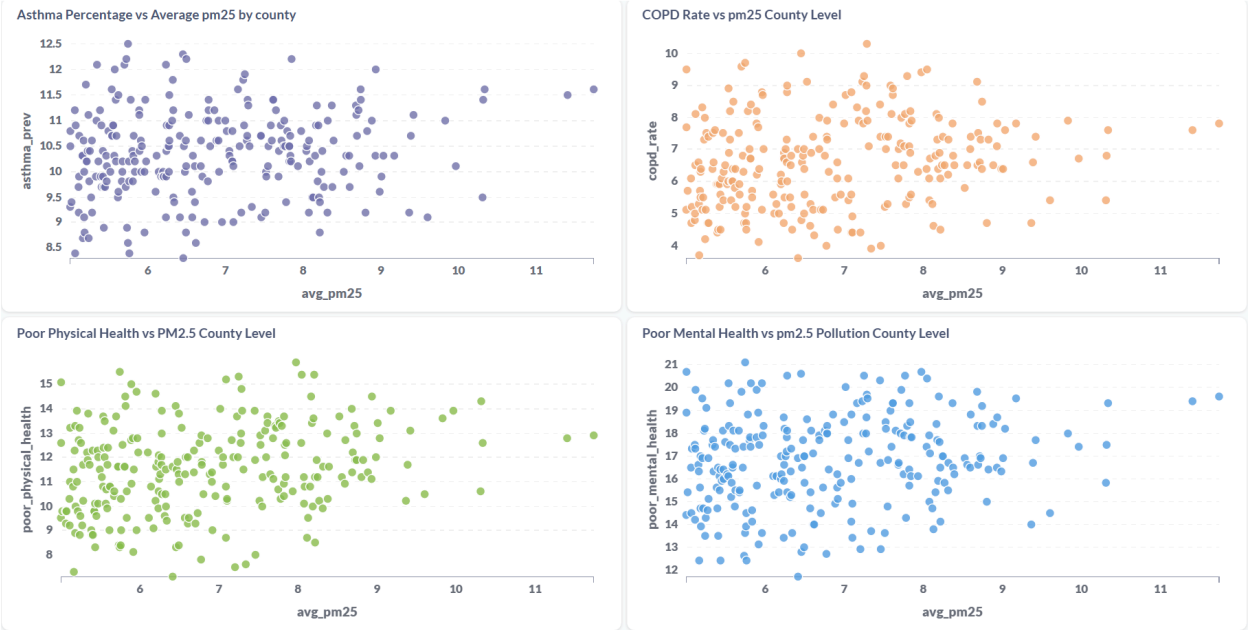


The line chart above shows average PM2.5 Levels across different poverty rate ranges. Counties with lower and average poverty percentages show similar PM2.5 exposure, which is higher than the average PM2.5 exposure shown in counties with high poverty percentages. This suggests that pollution exposure occurs across multiple ranges of economic conditions, rather than to just low income areas.

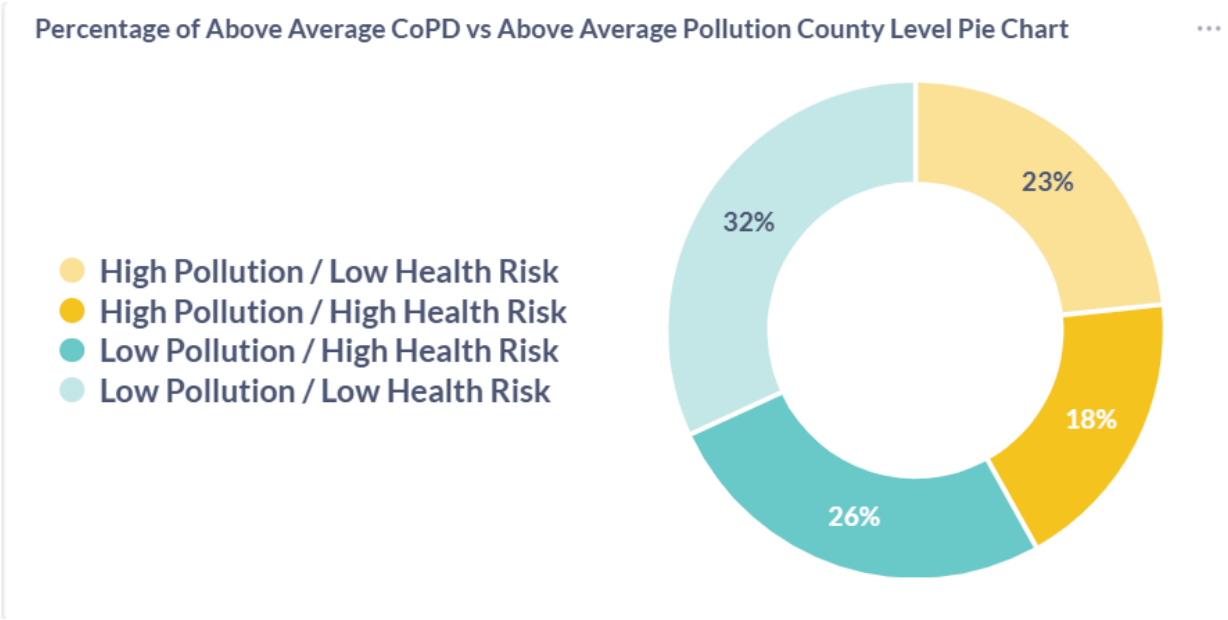
Community Health Outcomes Trends vs Pollution Exposure

Health Outcomes Included the Following:

- COPD Prevalence (%) – Long-term respiratory disease often linked to inhaled pollutants.
- Asthma Prevalence (%) – Chronic respiratory condition that may be aggravated by air quality.
- Poor Physical Health Prevalence (%) – Adults reporting 14+ days per month of poor physical health.
- Poor Mental Health Prevalence (%) – Adults reporting 14+ days per month of poor mental health.



The four scatter plots above compare PM2.5 Concentrations with Asthma, COPD, Poor Physical Health, and Poor Mental Health Rates to highlight a relationship between Pollution and health outcomes. Though the patterns vary for each condition, a higher PM2.5 level correlates loosely to a slightly Higher COPD, Asthma, Poor Physical Health, and Poor Mental Health. These visuals help identify counties where pollution may be correlated to a higher prevalence of chronic health concerns and health levels.



This pie chart helps show the percentage of counties that fall into the following subgroups: high pollution with high health risk, high pollution with lower health risk, low pollution with high health risk, and low pollution with lower risk. These subgroups show the percentage of counties that face both environmental and health concerns and see the scale of counties in the data set that need intervention for both concerns.

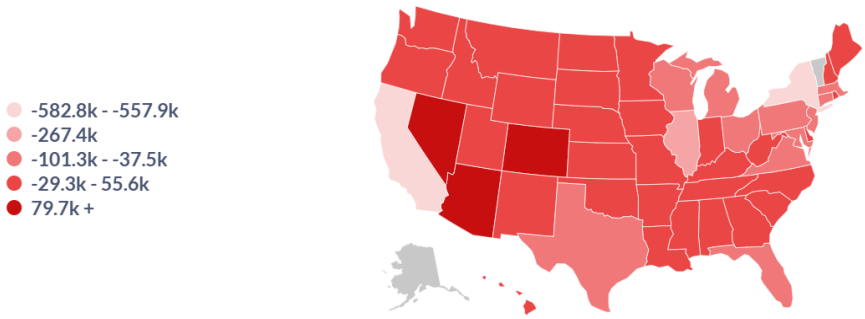
Migration Trends vs Environmental Factors

AQI vs Net Migration Table 2015-2021 Normalized

county_fips	state	countyname	pollution_year	Median_AQI	net_migration	Pct_Unhealthy
04013	arizona	Maricopa County Total Migration-US and Foreign	2,021	122	6,982	0.42
06037	california	Los Angeles County Total Migration-US and Foreign	2,020	85	-70,554	0.33
06003	california	Alpine County Total Migration-US and Foreign	2,015	102	-6	0.52
04013	arizona	Maricopa County Total Migration-US and Foreign	2,020	112	13,715	0.34
06037	california	Los Angeles County Total Migration-US and Foreign	2,021	79	-52,940	0.26
06071	california	San Bernardino County Total Migration-US and Foreign	2,020	93	-93	0.4
06037	california	Los Angeles County Total Migration-US and Foreign	2,016	77	-42,770	0.27
06037	california	Los Angeles County Total Migration-US and Foreign	2,019	72	-56,114	0.23
06037	california	Los Angeles County Total Migration-US and Foreign	2,017	79	-33,770	0.3
06037	california	Los Angeles County Total Migration-US and Foreign	2,018	78	-34,539	0.29

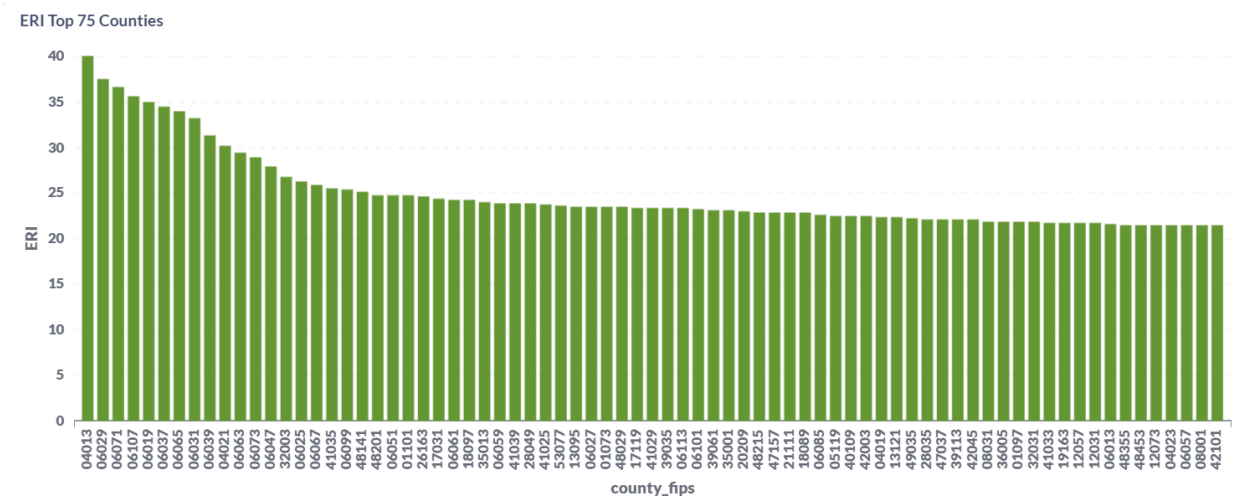
The table above highlights counties with a high average PM2.5 Concentration, Median AQI (Air Quality Index) Score, and Percentage of Unhealthy Days in a year that also suffer from a high negative net migration trend. A negative net migration value indicates residents leaving their current county. The counties highlighted in red represent priority communities where nonprofits can focus on advocacy, health programs, and policy efforts to improve the air quality and address the impacted population.

Net Migration Heatmap

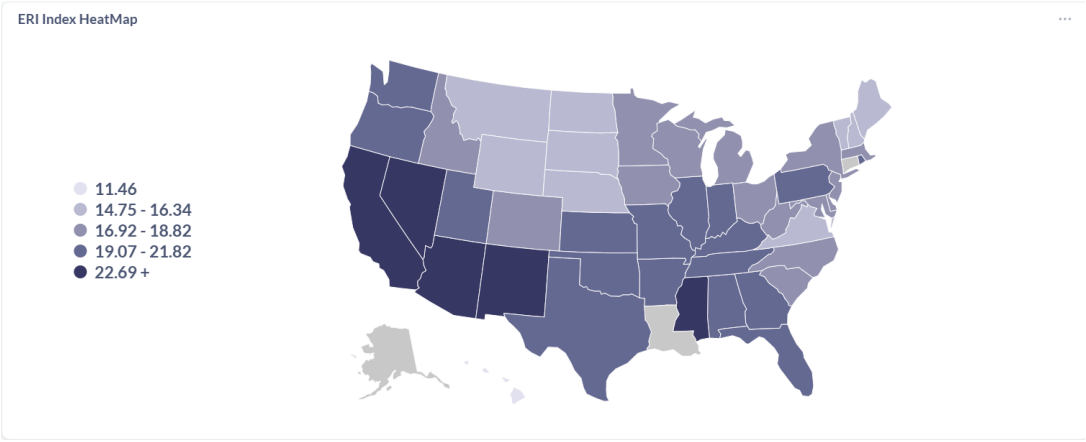


The Heat Map showcases the table data on a broader scale (State Level), displaying the Net Migration by state. Nonprofits looking to expand outreach into other states could view states with a negative net migration.

Environmental Risk Index (ERI): Identifying Vulnerable Counties for Nonprofit Intervention



This bar chart visualizes ERI values for the highest ranking counties and illustrates how risk levels fluctuate across impacted regions. This bar chart also showcases County Fips Codes, showing that these counties showcasing that Environmental risk is a concern throughout the United States and not just in a specific region.



The Heat Map showcases the ERI Average Values across each state, displaying the ERI Average by state. Nonprofits looking to expand outreach into other states could view states with a higher ERI Average.

Deliverables

- **Python Cleaning Scripts:**
  - clean_aqi_data_dynamic.py
  - clean_cdc_places_county.py
  - clean_conc_monitor.py
  - clean_irs_migration.py
  - clean_median_income.py
  - clean_poverty.py
- **SQL Create Table Scripts:**
  - create_air_quality_concentration.sql
  - create_aqi_county_data.sql
  - create_irs_migration.sql
  - create_median_income_2022.sql
  - create_places_county_health.sql
  - create_poverty_rates_2023.sql
- **NORP GitHub Code Additions:**
  - metabase_create_and_insert.txt: includes all code changes to create_tables.py and insert_into_tables.py
- **Metabase Dashboard:**
  - Live Dashboard on Metabase, accessible through http://130.207.3.31/norpmetabase/dashboard/44-pollution-exposure-risk-dashboard-environment-health-and-community-analysis?tab=60-pollution

All code the team produced can be found at our GitHub Repository: https://github.com/MirzaZuhayr/NORP-Pollution-Analysis

Skills Learned

Throughout the development of this project, we gained several important meta skills that extend beyond the technical implementation of data cleaning, integration, and analysis. These skills strengthened our ability to approach open-ended problems, validate assumptions, and recognize broader trends within complex datasets.

Applying Theory in Practice and Checking Assumptions

A significant component of the project involved taking conceptual knowledge about environmental indicators, socioeconomic metrics, and public health statistics and transforming it into a practical data integration workflow. Early in the process, we made several assumptions about how datasets would align across years, formats, and county identifiers. Implementing the pipeline required continually testing and correcting these assumptions, particularly around data completeness and pollutant categorization. This process developed our ability to treat theoretical expectations as hypotheses that must be verified through investigation and iterative fact checking. We learned to no longer make gross assumptions without proper fact checking supported by concrete evidence.

Data Engineering

Integrating data from the EPA, IRS, USDA, and CDC reinforced the importance of data engineering practices. Each dataset came with its own structural challenges such as inconsistent column naming, missing observations, or variations in geographic labeling. Through iterative refinement of our cleaning scripts, we learned to break down the larger integration task into smaller, more manageable components. This strengthened our skill in constructing reliable data pipelines that can scale across multiple years and data sources.

Trend and Pattern Recognition Across Domains

Analyzing the combined datasets taught us to identify high-level trends spanning environmental, economic, health, and demographic domains. For example, we

observed relationships between PM2.5 concentration and socioeconomic disadvantage, between AQI and health outcomes, and between pollution levels and county-level net migration. Recognizing these patterns required generalizing across datasets while also understanding domain-specific nuances. This allowed us to develop an advanced skill in identifying cross-domain trends rather than treating each dataset in isolation.

Specialization and Domain Sensitivity

Working with environmental data required not only programming skills but also an understanding of the difference between AQI and concentration metrics, as well as distinctions between pollutant types, sampling durations, and health-relevant thresholds. Learning when to specialize (e.g., focusing specifically on PM2.5 due to its established role in health risk) and when to generalize (e.g., consolidating indicators for building composite county risk scores) helped us refine the analytical framing of the project.

Distribution and Concentration Awareness

A recurring theme in the analysis involved examining how environmental and socioeconomic burdens are distributed across counties. Visualizations of PM2.5, poverty rates, migration flows, and health outcomes revealed strong geographic concentrations of risk, often aligning with rural, low-income, or industrial regions. Recognizing these patterns strengthened our ability to interpret distributional data and understand how spatial concentration of disadvantage informs nonprofit prioritization. This skill is particularly relevant to NORP's mission of supporting targeted, evidence-based action.

Visualization as a Reasoning Tool

Throughout the project, data visualization became essential not only for communicating findings but for reasoning through them. Scatterplots, heatmaps, line graphs, and trend comparisons allowed us to uncover correlations, identify outliers, and validate our interpretations. Developing effective visualizations reinforced how graphical analysis can reveal insights that are not immediately visible through tabular data alone.

Iterative Problem Decomposition and Flexibility

Working across multiple agencies' datasets required adapting to unexpected issues such as corrupted files, missing variables, and inconsistencies in county naming conventions. These challenges strengthened our flexibility and ability to pivot when initial approaches failed. Breaking down complex problems into smaller tasks and adapting workflows when needed became a core meta skill that improved the reliability of the overall project.

Milestone Chart

Weeks	Task
1-2	Research previous NORP projects and related work, brainstorming project ideas
3-4	Complete Project Checkpoint 1 Proposal, Find and collect relevant datasets.
5-6	Install and setup required software tools, setup development environments, refine scope with feedback adjustments and refine datasets
6-8	Refine dataset collection and perform data cleaning and transformation for relevant data with Python scripts
9-10	Integrate data sets into NORP Database and Metabase with SQL Create Table
10-12	Find relevant trends and perform thorough data analysis through SQL Queries
12-13	Create visualizations through SQL queries and map out insights/conclusions

References

1. **Pu, Calton, Mirae Kim, Teresa Derrick-Mills, and Lewis Fault.** “Challenges and Experiences in Data Integration to Support Research on Nonprofit Organizations.”
2. **De Silva, Dakshina G., et al.** “Tiebout Sorting and Toxic Releases.” *Environmental and Resource Economics*, vol. 87, 2024, pp. 2487–2520. https://doi.org/10.1007/s10640-024-00893-8.
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